CodeSet 8 - Ian Wyse - CMAN

1. There are 1024 nodes in the output layer, each with a 50Hz average fire rate. The average number packets per second is then $1024 \cdot 50 = 50$ Kpackets/s.

2. Each packet is 20 bits, so the bandwidth is 50 Kpacket/s $\cdot$ 20 bits/packet = 1000 Kbits/s. $\simeq$ 1 Mbit/s.

3. All three of SPI, I^2C, and AXI4-Lite have sufficient bandwidth for the average packet throughput. Both I^2C and SPI are quite simple, but I^2C would probably more appropriate since it better manages multiple endpoints.

4. If 25% of neurons fire within 1ms, the instantaneous bandwidth is (1024 neurons $\cdot$ .25 packets/neuron $\cdot$ 20 bits/packet) / 1ms = 5120 bits/1ms = 5,120,000 b/s = 5 Mb/s. This is too much bandwidth for I^2C, but managable for SPI and AXI stream. If I^2C were used, buffering would be required, with a depth of at least 2Mb to absorb the overhead that the stream cannot. It would be safer to provide enough buffering to absorb the entire burst, which would require 5 Mb. The ratio of burst to average bandwidth is 5 Mb/s / 1 Mb/s $\approx$ 5.

5. A frame based readout would sample 1024 neurons every 1ms, producing 1024 bits or 1Kb. The bandwidth is then 1Kb/1ms = 1000 Kb = 1 Mb. This means that at 50Hz average fire rate, the throughput is the same, based on previous calculations. The crossover frequency is then 50Hz, meaning that AER is superior at less than 50Hz, with frame based readout at greater than 50Hz. A benefit of frame based readout is that data throughput is consistent regardless of neuron output behavior, so buffering is not required. Therefore for equal bandwidth cases such as this one, it makes sense to choose frame based over AER, assuming that there is no computation overhead to frame based readout compared to AER.